

CDS522 Text and Speech Analytics

Assignment 2

Report

Title: An Applied Study of Speech Emotion Classification Using Mel Frequency Cepstral Coefficient Features and Classical Machine Learning Models

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Abstract

Speech emotion recognition is a crucial process in building an emotion-sensitive dialogue system, which is practically useful for automated call centre dialogue system, teaching assistant robot and psychological therapy robot. Speech emotion recognition can be performed using classical machine learning models based on Mel frequency cepstral coefficient (MFCC) features. This project uses MFCC computed from a dataset that has 11318 short speech recordings with six emotion labels, namely anger, disgust, fear, happy, neutral and sad, to train the machine learning models. In this project, 11 machine learning algorithms, namely decision tree, support vector machine with linear kernel, support vector machine with polynomial kernel, support vector machine with radial basis function kernel, support vector machine with sigmoid kernel, k nearest neighbours, Gaussian naïve Bayes, Bernoulli naïve Bayes, logistic regression, multi-layer perceptron and random forest, are trained using mean of MFCC and compared based on the accuracy of speech emotion recognition. The three best models, namely k nearest neighbours, support vector machine with polynomial kernel and multi-layer perceptron are separately trained with mean and standard deviation of MFCC to investigate the difference in accuracy compared to mean of MFCC. It is found all three models has improved in its accuracy for speech emotion recognition task. The universal best performing model in this project is support vector machine with polynomial kernel trained with mean and standard deviation of MFCC, achieving accuracy of 0.6144.

Introduction

Speech is a very fundamental method of communication between human. Human speech naturally carries emotion. People can identify the emotion state of the speaker when they hear the speech. People can catch the emotion even if they don't understand the language. This is because people can identify emotion based solely on the intonation and loudness of the speech. This forms the basis for speech emotion recognition, which trains computer to identify speaker emotion from speech acoustic features per se, regardless of the content of the speech.

Speech emotion recognition has a broad range of use cases. One of the use cases is dialogue system that is able to recognize the emotion of the user when the user is speaking and give appropriate response based on the user emotion. This is useful for automated call centre

dialogue system, teaching assistant robot and psychological therapy robot. A good speech emotion recognition system can effectively improve the user experience of a dialogue system.

There are many types of features that are useful for speech emotion recognition, such as spectrogram and Mel frequency cepstral coefficient (MFCC). The scope of this project specifically focuses on MFCC features. MFCC contains information of the formant, which corresponds to the shape of vocal tract when producing sound. When people have different emotions, their vocal tract shape and the formant produced are slightly different. MFCC is also sensitive to energy of speech. When the emotion of speaker is different, the energy of the speech is different. For example, based on the research of Kumar & Iqbal (2023), when the speaker is feeling angry, joy or happy, the energy or intensity of the speech is high, whereas when the speaker is feeling disgust or fear, the energy or intensity of the speech is low.

The first objective of this project is to find the classical machine learning algorithms that is able to produce good speech emotion recognition model based on MFCC features. Mean of MFCC is used in this project because it is commonly used in the literature. The project compares among decision tree, support vector machine (linear kernel, polynomial kernel, radial basis function kernel and sigmoid kernel), k nearest neighbours, naïve Bayes (Gaussian and Bernoulli), logistic regression, multi-layer perceptron and random forest.

The second objective of this project is to study whether mean and standard deviation of MFCC can yield better performance for speech emotion recognition than mean of MFCC. It is hypothesized that more statistical measures of MFCC yields better performance than less statistical measures of MFCC. This project specifically uses the top three algorithms discovered in the first objective to test the effect of adding standard deviation of MFCC on the performance of speech emotion recognition.

Literature review

This project focuses on speech emotion recognition using MFCC and classical machine learning algorithms. So, previous works that used MFCC and machine learning to perform speech emotion recognition is reviewed.

Ancilin & Milton (2021) compared features of linear prediction cepstral coefficient (LPCC), log frequency power coefficient (LFPC), MFCC and Mel frequency magnitude coefficient (MFMC) based on the performance on speech emotion recognition task using support vector machine classifier. In the research, various statistical measures, such as mean, median, standard deviation, interquartile range and range, are used to extract the feature from LPCC, LFPC, MFCC and MFMC (Ancilin & Milton, 2021). The features are compared across different speech datasets and MFMC achieved the highest performance, with accuracy of 81.50%, 64.31%, 75.63%, 73.30% and 56.41% and 95.25% for six different datasets. (Ancilin & Milton, 2021) Ancilin & Milton (2021) also experimented 12 coefficients, 24 coefficients and 30 coefficients for the features and found out that the higher order coefficient matrices has higher the performance.

Mishra, Warule, & Deb (2024) compared mean MFCC, MFCC-based spectral entropy and MFCC-based approximate entropy for speech emotion recognition using neural network classifier. Various combinations of the three feature representations are experimented on multiple datasets and it is concluded that the combination of mean MFCC and MFCC-based spectral entropy achieved the best performance, with accuracy of 87.48%, 75.9% and 79.64% for three different datasets (Mishra, Warule, & Deb, 2024). Mishra, Warule, & Deb (2024) also tested different order of coefficient matrix (12 coefficients, 20 coefficients and 30 coefficients) and found that higher order coefficient matrices has better performance.

Mustafa, et al. (2021) used support vector machine to classify speech emotions based on MFCC features extracted from the speech. The models in the research achieved accuracy of 97% for Toronto University dataset and 86% for IEMOCAP dataset (Mustafa, et al., 2021).

Kumar & Iqbal (2023) proposed a decision-tree-based algorithm named BOAT and association rule mining to recognize speech emotions based on MFCC. In the research, mutual information is used to perform feature selection on the original features before the selected features are fed into the classifier (Kumar & Iqbal, 2023). The model proposed by the research achieved accuracy of 85.95% and clearly outperformed the baseline model (support vector machine) in speech emotion recognition task (Kumar & Iqbal, 2023).

Nurdiawan, Kurnia, Sudrajat, & Pratama (2025) compared support vector machine, k nearest neighbours, decision tree and an ensemble of these three models for speech emotion recognition. The feature used in the research is mean and standard deviation of 13-coefficient

MFCC (Nurdiawan, Kurnia, Sudrajat, & Pratama, 2025). The results showed that the ensemble model outperformed the single models, achieving accuracy of 80.43%, while k nearest neighbours and decision tree are the best performing single model (both has accuracy of 78.26%), followed by support vector machine (accuracy of 52.17%) (Nurdiawan, Kurnia, Sudrajat, & Pratama, 2025).

The literatures that are reviewed in this project are summarised in Table 1 in terms of the machine learning algorithm used, number of MFCC coefficients used and the variants of MFCC used.

Table 1 Summary of literature

Literature	Machine learning algorithm	Number of MFCC coefficients	Variants of MFCC
(Ancilin & Milton, 2021)	Support vector machine	12, 24, 30	Various statistical measures of MFCC (mean, standard deviation, interquartile range, range, etc.)
(Mishra, Warule, & Deb, 2024)	Neural network	12, 20, 30	Mean of MFCC, MFCC-based spectral entropy, MFCC-based approximate entropy
(Mustafa, et al., 2021)	Support vector machine	Not mentioned	Not mentioned
(Kumar & Iqbal, 2023)	Support vector machine, BOAT (decision-tree-based model) + association rule mining	Not mentioned	Not mentioned
(Nurdiawan, Kurnia, Sudrajat, & Pratama, 2025)	Support vector machine, k nearest neighbours, decision tree, ensemble of these three models	13	Mean and standard deviation of MFCC

As shown in Table 1, the machine learning algorithms that are typically used for speech emotion recognition using MFCC features are support vector machine, decision tree, k nearest neighbours and neural network (multi-layer perceptron). In the literatures, naïve Bayes, logistic regression and random forest are not explored. This is the research gap found from the literature review. So, to fill the research gap, in this project, all aforementioned machine learning algorithms, namely support vector machine, decision tree, k nearest neighbours, multi-layer perceptron, naïve Bayes, logistic regression and random forest, will be explored.

As shown in Table 1, various variants of MFCC are explored by the researchers. There are 3 out of 5 literatures use mean of MFCC in their research and 2 out of 5 literatures use standard deviation of MFCC. So, in this project, the most popular statistical measure of MFCC, i.e. mean, will be used to generate the features for all models in the first round. After that, to investigate the effect of more statistical measures on the performance of speech emotion recognition, the three best models in the first round will be further tested with features of mean MFCC and the second popular statistical measures of MFCC, i.e. standard deviation, in the second round.

As shown in Table 1, most literatures (3 out of 5) use 12 or 13 coefficients for MFCC. So, 13-coefficient MFCC will be used as the features in the first and the second round.

Methodology

This project will work on a speech emotion recognition problem. The main goal of speech recognition problem is to classify a given speech into the correct emotion category. So, this can be formulated into a supervised machine learning problem, or more specific, a multi-class classification problem.

The dataset used in this project is a speech dataset with 11318 instances and labelled with 6 emotions, namely anger, disgust, fear, happy, neutral and sad. The dataset originates from Kaggle and is uploaded by Abozaid (2023).

The dataset is balanced because the number of instances of each label ranges from 1703 to 1923. So, in this project, accuracy is an appropriate metric to measure the performance of the machine learning models on speech emotion recognition.

The aim of this project is to find the best machine learning algorithm for speech emotion recognition using MFCC features of a speech. Besides that, this project also intends to find whether using mean and standard deviation of MFCC can improve the performance of the model on speech recognition task compared to using only the mean of MFCC.

The general flow of experiments to achieve the aforementioned two purpose in this project is depicted in Figure 1.

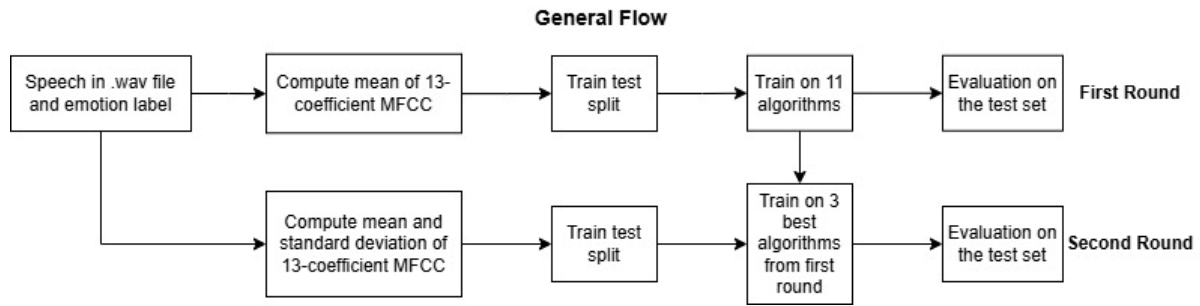


Figure 1 General flow of experiments

As shown in Figure 1, each speech file is loaded using the Librosa library and the 13-coefficient MFCC of the speech is computed using the same library. The MFCC has a sequence of feature vectors, where each feature vector represents a period of time in the speech and there are 13 coefficients for each feature vector. In the first round, the mean of each coefficient is computed using the NumPy library, whereas in the second round the mean and the standard deviation of each coefficient is computed using the same library. These are used as the input features for the machine learning algorithms. In other words, in the first round, there are 13 features, whereas in the second round, there are 26 features. The hypothesis is the mean + standard deviation (26 features) will outperform the mean (13 features). The label used in training is the pre-defined 6-category emotion label in the dataset.

After the features are generated and the label are fixed, the next step is performing train test split. The test size is 0.2, which means 20% of the dataset are set as the test set. The test set will not be used in the training process and will be used to evaluate the performance of the model. In the first round, 11 machine learning algorithms, namely decision tree, support vector machine with linear kernel, support vector machine with polynomial kernel, support vector machine with radial basis function kernel, support vector machine with sigmoid kernel, k nearest neighbours, Gaussian naïve Bayes, Bernoulli naïve Bayes, logistic regression, multi-layer perceptron and random forest, are trained and evaluated. The mean of MFCC is used as the features. In the second round, the three best performing algorithms in the first round are trained using the mean + standard deviation of MFCC features and evaluated. Since the class distribution is balanced, accuracy is used to evaluate the performance of the models.

The software used in this project is Jupyter notebook. All code is run locally on a personal laptop.

Results

In the first round, the hyperparameter of each algorithm is tuned using grid search using the training set. Then, the best hyperparameters are used to train the speech emotion recognition model. Every model is evaluated by testing it using the same test set. Table 2 lists the test accuracies of first round, i.e. using mean of MFCC (13 features), as well as the best parameters of each model.

Table 2 Test accuracy of each model using mean of MFCC

Algorithm	Best hyperparameters (after tuning)	Test accuracy
Decision tree	criterion: entropy, splitter: best, max_depth: 10, min_samples_split: 20, min_samples_leaf: 20, min_impurity_decrease: 0.01, class_weight: balanced	0.4170
Support vector machine with linear kernel	C: 1, max_iter: 10000	0.1855
Support vector machine with polynomial kernel	C: 10, degree: 6, coef0: 1.0	0.5269
Support vector machine with radial basis function kernel	C: 10	0.4620
Support vector machine with sigmoid kernel	C: 0.1, coef0: 0.0	0.1603
K nearest neighbours	n_neighbors: 25, p: 1, weights: distance	0.5411
Gaussian naïve Bayes	var_smoothing: 0.01	0.3547
Bernoulli naïve Bayes	alpha: 0.1, binarize: 0.2	0.3065
Logistic regression	penalty: l2, class_weight: None, solver: newton-cholesky, max_iter: 5000	0.3883
Multi-layer perceptron	hidden_layer_sizes: (20, 30, 15, 8), activation: relu, solver: adam, max_iter: 500	0.5000
Random forest	n_estimators: 100, criterion: entropy, max_depth: 30, min_samples_split: 50, min_samples_leaf: 20, min_impurity_decrease: 0.01	0.4806

Figure 2 illustrates the test accuracies of each speech recognition model using mean of MFCC.

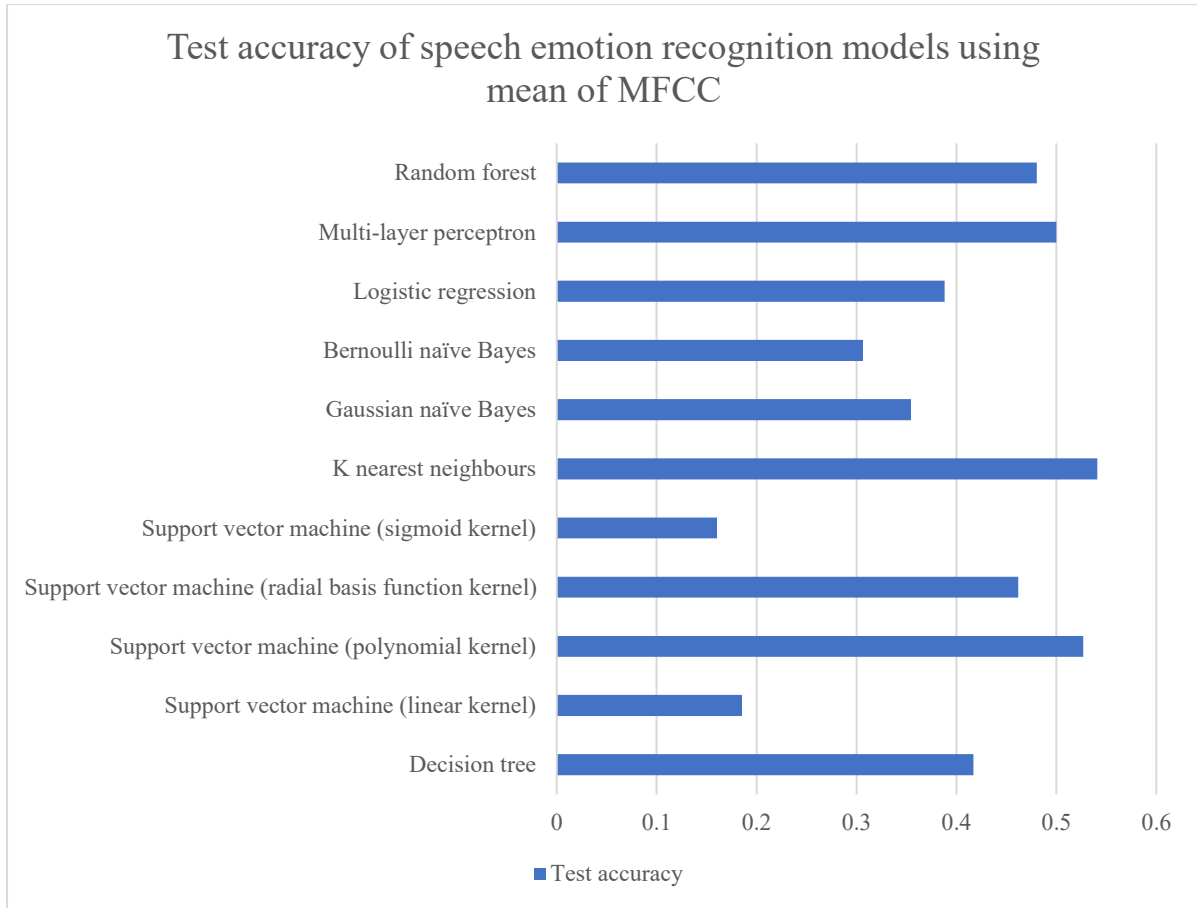


Figure 2 Test accuracy of each model using mean of MFCC

Based on Figure 2, it is clear that the three best performing models are k nearest neighbours, support vector machine with polynomial kernel and multi-layer perceptron. So, these three algorithms are used in the second round that further investigates the improvements in the performance of the models by adding extra features, which is the standard deviation of MFCC.

In the second round, to ensure fairness, the same hyperparameters are used to train the models with the expanded feature sets (mean + standard deviation of MFCC). The models are evaluated by testing on the same test set used in the first round. Table 3 compares the test accuracy of the models when using the mean of MFCC and when using the mean and standard deviation of MFCC. In Table 3, the improvement in test accuracy after adding standard deviation of MFCC into the feature set (which initially only has mean of MFCC) is also reported.

Table 3 Comparison of test accuracy of models using mean of MFCC and mean + standard deviation of MFCC

Algorithm	Test accuracy when using mean of MFCC	Test accuracy when using mean + standard deviation of MFCC	Improvement in test accuracy
Support vector machine with polynomial kernel	0.5269	0.6144	0.0875
K nearest neighbours	0.5411	0.5813	0.0402
Multi-layer perceptron	0.5000	0.5605	0.0605

Figure 3 illustrates the test accuracies of the three models when using mean of MFCC and when using mean + standard deviation of MFCC.

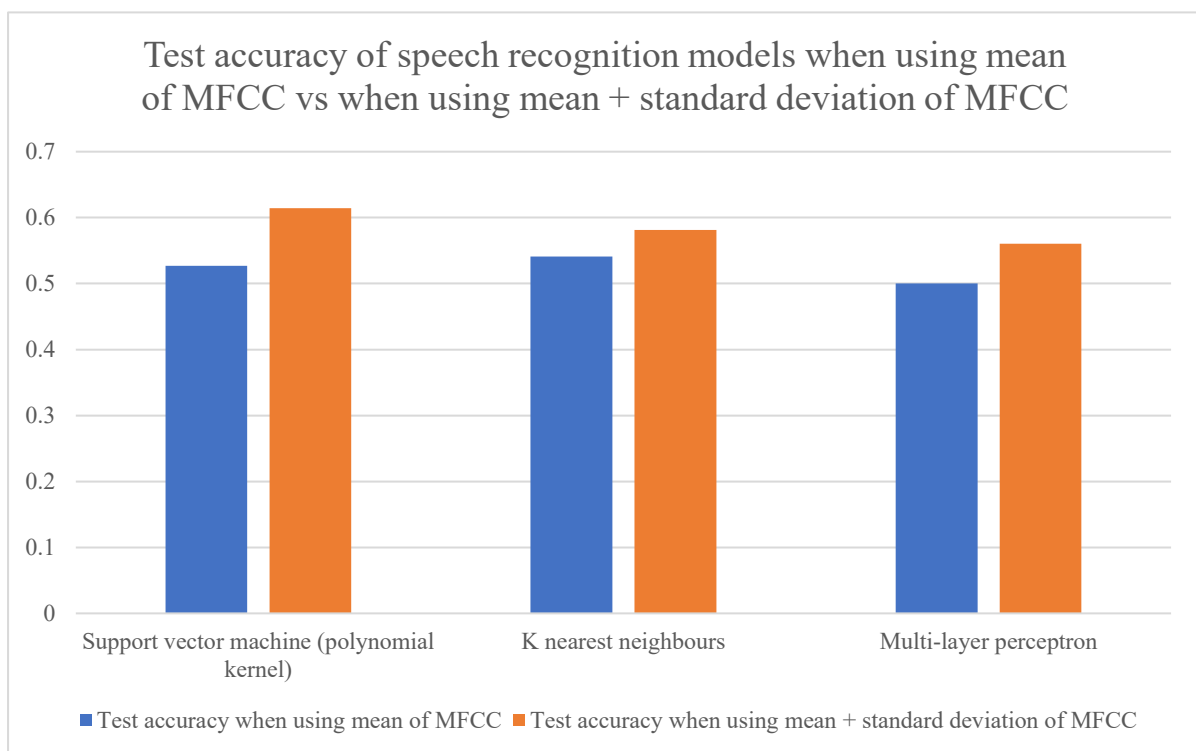


Figure 3 Test accuracy of each model when using mean of MFCC vs when using mean + standard deviation of MFCC

Figure 4 is the confusion matrix of the best performing model, the support vector machine with polynomial kernel model, with mean and standard deviation of MFCC.

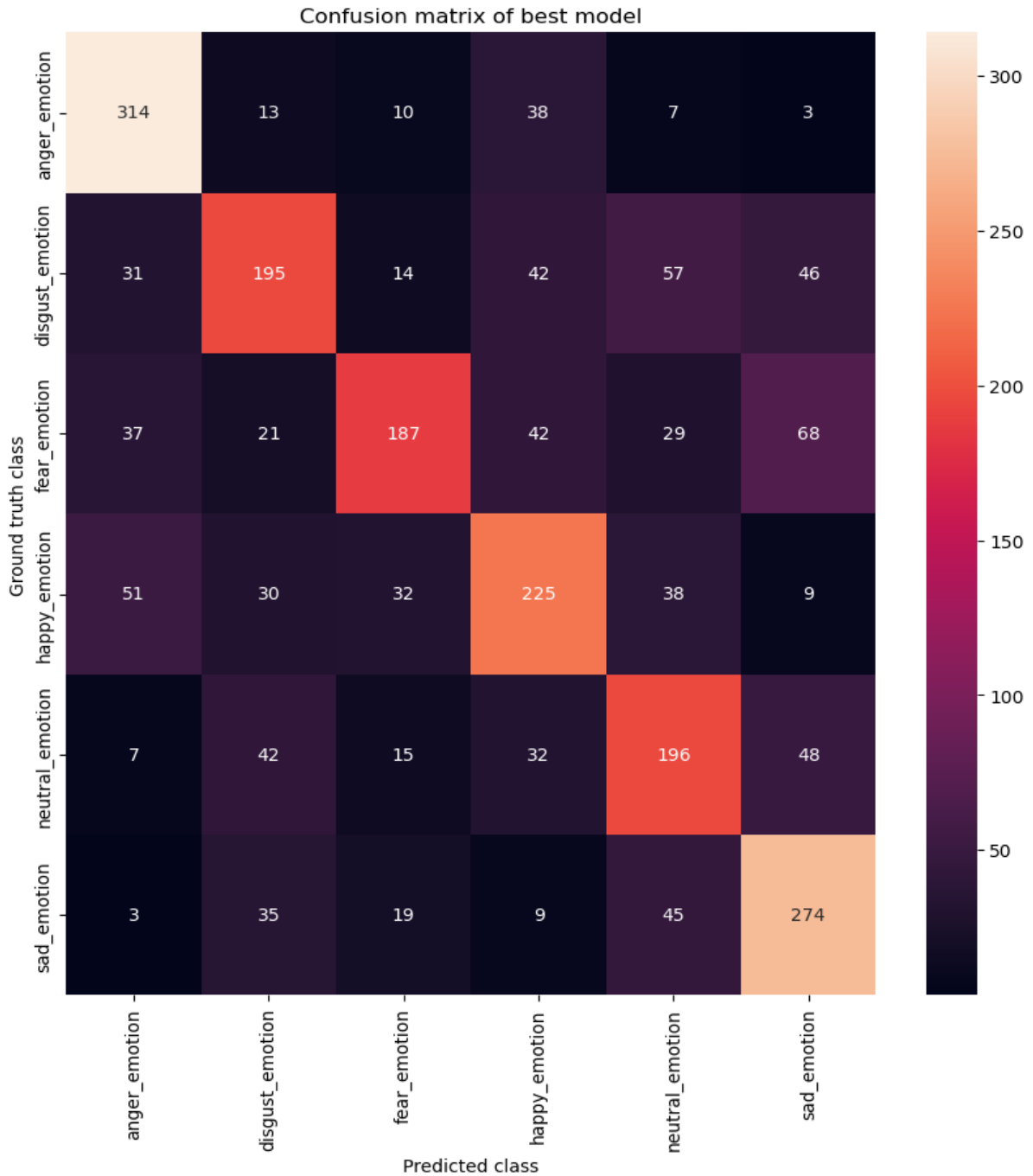


Figure 4 Confusion matrix of best model

Discussion

In the first round, 11 machine learning algorithms are compared for speech emotion recognition task using mean of MFCC as the input features. If the instances are labelled at random, the probability of each label is 1 divided by 6, which is 0.1667. The accuracy of all models, except for support vector machine with sigmoid kernel, surpass the random probability

of 0.1667. This means that all models except support vector machine with sigmoid kernel is better than classifying the speech randomly. In other words, all models except support vector machine with sigmoid kernel has learned something from the training process. So, it can be said that, although it is not very good, supervised learning with classical machine learning algorithms has the potential to tackle speech emotion recognition problem to certain extent. At the very least, it is better than label randomly.

In the first round, it is found that among the 11 machine learning models, the best performing model is k nearest neighbour, with test accuracy of 0.5411, followed by support vector machine with polynomial kernel, with test accuracy of 0.5269, and multi-layer perceptron, with test accuracy of 0.5000. It is important to note that the top three algorithms are algorithms that are able to model non-linear relationships. So, it can be said that algorithms that are able to model non-linear relationships perform better for speech emotion recognition task. This might be because of the relationship between MFCC and emotion of speech is complex and non-linear.

In the second round, the machine learning algorithms that yield the top three models in the first round are used to train a speech emotion recognition model using mean and standard deviation of MFCC. It is found that the performance of the models has improved significantly. The improvement of accuracy ranges from 0.0402 to 0.0875. So, it can be said that using mean and standard deviation of MFCC as the features results in better performance on speech emotion recognition compared using only mean of MFCC. This is because standard deviation gives additional information for the models to analyse the MFCC in a different perspective.

In the second round, it is also important to note that the support vector machine with polynomial kernel model overtakes the k nearest neighbours model to become the best performing model after adding in standard deviation of MFCC into the feature set. The test accuracy of support vector machine is 0.6144, which is the best, and it is followed by k nearest neighbours, with test accuracy of 0.5813 and multi-layer perceptron, with test accuracy of 0.5605. So, the support vector machine with polynomial kernel model with mean and standard deviation of MFCC as the features, achieves the universal best performance in this project.

Based on the confusion matrix of the universal best performing model in this project in Figure 4, the model mostly identified the labels correctly. This is because the cells of the correct

prediction (the diagonal of confusion matrix) has clearly higher counts (3-digit numbers) compared to the cells of the incorrect prediction (2-digit number and 1-digit number).

The model is most confused when differentiating fear emotion and sad emotion because the count of instances that are actually fear emotion but predicted as sad emotion is the highest (68 instances). Based on the research done by Kumar & Iqbal (2023), the intensity of fear emotion is low, whereas the intensity of sad emotion is inflecting downwards. So, there are similarities in intensity of fear emotion and sad emotion, which causes the model to mix up these two emotions.

The model is most not confused when differentiating anger emotion and sad emotion because both the count of instances that are actually anger emotion but predicted as sad emotion or actually sad emotion but predicted as anger emotion are the lowest (3). Based on the research done by Kumar & Iqbal (2023), the intensity of anger emotion is high, whereas the intensity of sad emotion is inflecting downwards. So, there is a huge difference between anger emotion and sad emotion, which helps the model to differentiate these two emotions correctly.

Table 4 compares the accuracy of the best models from the literature with the best model from this project.

Table 4 Comparison of the best models from literature and this project

Source	Algorithm for best model	Feature representation for best model	Accuracy of best model
(Ancilin & Milton, 2021)	Support vector machine	Mel frequency magnitude coefficient	0.9525
(Mishra, Warule, & Deb, 2024)	Neural network	Mean of MFCC + MFCC-based spectral entropy	0.8748
(Mustafa, et al., 2021)	Support vector machine	MFCC	0.9700
(Kumar & Iqbal, 2023)	Decision-tree-based BOAT + association rule mining	MFCC	0.8595
(Nurdiawan, Kurnia, Sudrajat, & Pratama, 2025)	Ensemble of decision tree, support vector machine and k nearest neighbours,	Mean + standard deviation of MFCC	0.8043
This project	Support vector machine with polynomial kernel	Mean + standard deviation of MFCC	0.6144

Based on Table 4, comparing to the best models from the literature, the best model from this project has the lowest accuracy. It is also important to note that this project is not using the same dataset with the literature, so the comparison is actually not a fair comparison. Nonetheless, the performance of the best model in this project is actually not very good, since its accuracy is only 0.6144. So, in future work, the model has to be improved in order to achieve more promising performance.

One way to improve performance is adding more statistical measures of MFCC to the feature. This is because it has been proven from this project that features of mean + standard deviation of MFCC yields better performance than mean of MFCC. So, it is legitimate to hypothesize that adding more statistical measures of MFCC, such as range, interquartile range, maximum and minimum, to the feature can possibly improve the performance of the model.

Conclusion

As a conclusion, the first objective of this project is achieved. 11 classical machine learning algorithms are compared based on the performance on speech emotion recognition task based on mean of MFCC. It is found that the top three models are trained using k nearest neighbours, support vector machine with polynomial kernel and multi-layer perceptron, with test accuracy of 0.5411, 0.5269 and 0.5000 respectively.

The second objective is also achieved. The top three models for mean of MFCC are re-trained by using mean and standard deviation of MFCC. It is found that the hypothesis of using mean and standard deviation of MFCC yields a better performance than mean of MFCC is true. This is because the test accuracy of k nearest neighbours, support vector machine with polynomial kernel and multi-layer perceptron has increased for 0.0402, 0.0875 and 0.0605 respectively. It is meaningful to note that the support vector machine with polynomial kernel has surpassed k nearest neighbours to become the best performing model after adding standard deviation of MFCC to the feature set. So, the support vector machine with polynomial kernel model is the universal best performing model in this project.

One possible future improvement is trying to add more statistical measures of MFCC. This direction is proven valid in this project by comparing mean + standard deviation of MFCC and mean of MFCC. However, it is still unknown whether further adding more statistical

features, such as median, maximum, minimum, range and interquartile range, of MFCC will further improve the performance for speech emotion recognition. Although it is unknown, but it is a direction of improvement that has high potential given in this project two statistical measures outperform one statistical measure.

Recommendation

Based on the findings in this project, it is recommended to use support vector machine with polynomial kernel, k nearest neighbours and multi-layer perceptron to train models for speech emotion recognition. Besides that, it is also recommended to add more statistical measures of MFCC to the feature set to improve the performance of speech emotion recognition.

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